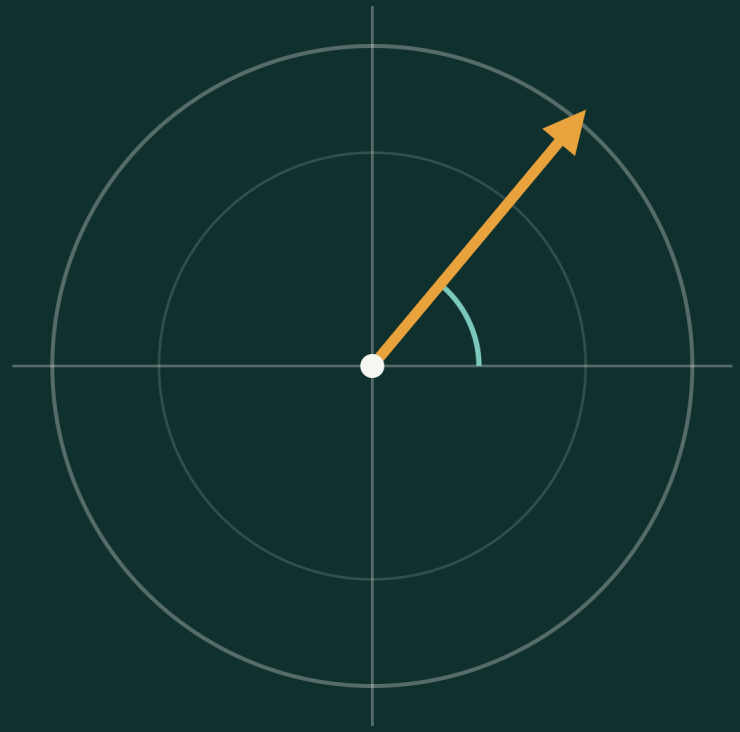
MATH AND THEORY FOR BEGINNERS

Quantum Computing from the Ground Up

From complex numbers to Shor's algorithm



Louis Libat - October 2026

THE COURSE

No analogies: how quantum computers actually work

“Qubits are 0 and 1 at the same time” is the usual answer, and it explains very little. We build the real picture step by step.

1
Mathematics
Complex numbers, matrices, inner products

2
A single qubit
Physics, measurement, Bloch sphere, gates, phase

3
Multiple qubits
Entanglement, teleportation, no-cloning, universal gates

4
Algorithms
Deutsch–Jozsa, Grover, QFT, RSA and Shor

5
A quantum world
Cryptography, error correction, advantage, myQLM code

PART 1 · MATHEMATICS

Imaginary and complex numbers

The problem

$$x^2 = 4 \;\Rightarrow\; x = \pm 2$$

$$x^2 = -4 \;\Rightarrow\; ?$$

A real number squared is never negative. The fix is to define

$$i = \sqrt{-1}, \qquad \sqrt{-4} = \pm 2i$$

A complex number = real part + imaginary part

$$z = a + ib, \qquad a, b \in \mathbb{R}$$

Add and subtract part by part:

$$(3 + 2i) + (1 - 4i) = 4 - 2i$$

Multiply, remembering $i^2 = -1$:

$$(2 + 3i)(1 - i) = 2 - 2i + 3i - 3i^2 = 5 + i$$

Conjugate, complex plane and magnitude

Complex conjugate: flip the imaginary sign

$$z = a + ib \;\Rightarrow\; z^* = a - ib$$

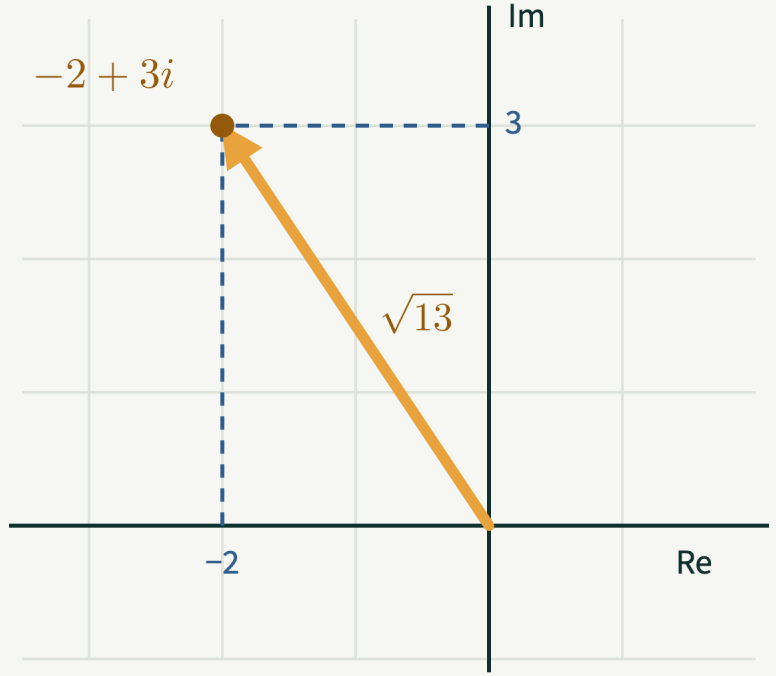
A number times its conjugate is always real:

$$(2 + 3i)(2 - 3i) = 4 + 9 = 13$$

Magnitude: distance from the origin

$$|a + ib| = \sqrt{a^2 + b^2}$$

$$|4 - 3i| = \sqrt{16 + 9} = 5$$



Exponential form: multiplying means rotating

Describe z by its length r and angle θ:

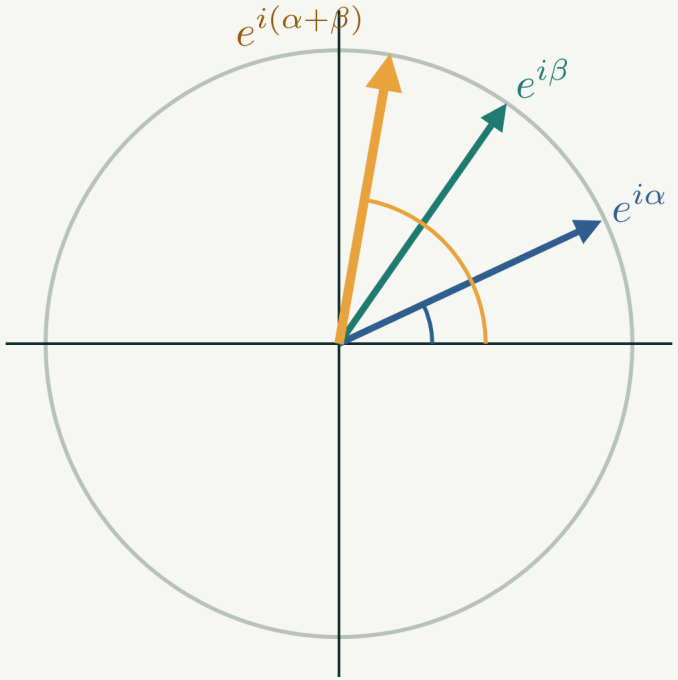
$$z = r(\cos\theta + i\sin\theta) = r\,e^{i\theta}$$

$$r = \sqrt{a^2 + b^2}, \qquad \theta = \arctan\frac{b}{a}$$

Example: $1 + i = \sqrt{2}\,e^{i\pi/4}$

$$e^{i\alpha} \cdot e^{i\beta} = e^{i(\alpha+\beta)}$$

On the unit circle, angles **add**: multiplication is a rotation. This is how phase will rotate qubits.



PART 1 · MATHEMATICS

Matrices

Shape, addition, scaling

$\begin{pmatrix} 1 & 4 & 2 \\ 0 & 3 & 5 \end{pmatrix}$ is 2 × 3: **m rows, n columns**

Add or subtract element by element, only for the same shape.

Scalar multiple: multiply every element.

Product: row · column

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$

Top-left entry: $1 \cdot 5 + 2 \cdot 7 = 19$

Defined only if the left matrix has as many **columns** as the right one has **rows**.

PART 1 · MATHEMATICS

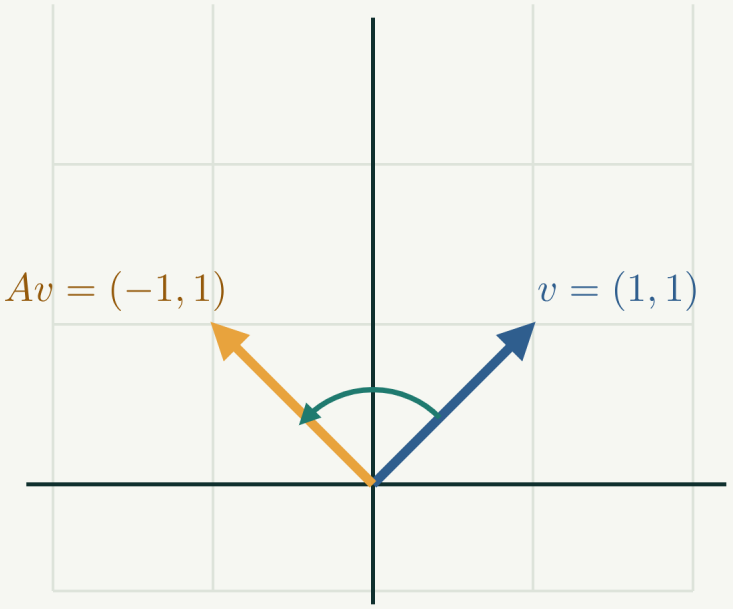
Matrices transform vectors

An n × 1 matrix is a **column vector**. A matrix times a vector gives another vector:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

This matrix rotates every vector by 90°.

> Quantum gates work exactly like this: a **matrix applied to the state vector**.



PART 1 · MATHEMATICS

Inverse, dagger, unitary and Hermitian

Identity

$AI = IA = A$

Ones on the diagonal, zeros elsewhere

Inverse

$AA^{-1} = I$

Apply A, then its inverse, and you are back

Dagger

$A^\dagger = (A^*)^T$

Conjugate every element, then swap rows and columns

Unitary $UU^\dagger = I$

U† is the inverse, so the gate is reversible. Equivalent tests:

$\langle Uu \,|\, Uv \rangle = \langle u \,|\, v \rangle, \qquad \|Uu\| = \|u\|$

or: its columns are orthonormal. Length is total probability, so **every gate is unitary**.

Hermitian $H = H^\dagger$

Equal to its own conjugate transpose; its eigenvalues are real.

X, Y, Z and the Hadamard gate are **both** unitary and Hermitian, so each is its own inverse:

$U = U^\dagger = U^{-1} \;\Rightarrow\; U^2 = I$

PART 1 · MATHEMATICS

Bra, ket and the inner product

Bra and inner product

A **bra** is the conjugate transpose of a ket: a row vector.

$$\langle\varphi| = |\varphi\rangle^{\dagger} = \begin{pmatrix} x_1^* & x_2^* & \cdots & x_n^* \end{pmatrix}$$

Row times column gives the inner product: “bra-ket” is a pun on *bracket*.

$$\langle\varphi|\psi\rangle = \sum_{i=1}^{n} x_i^*\, y_i$$

Norm and orthogonality

$$\|\psi\|^2 = \langle\psi|\psi\rangle = \sum_i |y_i|^2$$

Two states are **orthogonal** when their inner product is zero:

$$\langle 0|1\rangle = 0, \qquad \langle +|-\rangle = 0$$

Rules: swapping conjugates; a matrix moves across as its dagger.

$$\langle\psi|\varphi\rangle = \langle\varphi|\psi\rangle^*, \qquad \langle Au|v\rangle = \langle u|A^{\dagger}v\rangle$$

Eigenvectors and eigenvalues

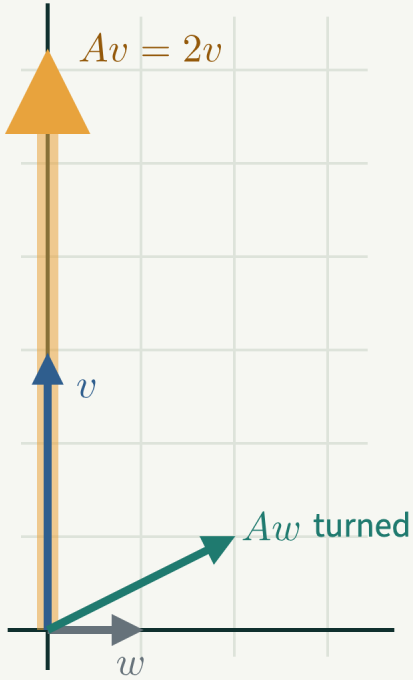
If a matrix only **stretches** a vector without turning it,

$$A\,v = \lambda\,v$$

then v is an **eigenvector** and λ an **eigenvalue**.

$$\begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 6 \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 3 \end{pmatrix}$$

Eigenvalue 2, eigenvector (0, 3). This idea drives **phase kickback** and **phase estimation** later.



PART 2 · A SINGLE QUBIT

A little quantum physics

Quantisation

$$\Delta E = n\,h\,\nu, \qquad h \approx 6.62 \times 10^{-34}\ \mathrm{J\,s}$$

Energy is exchanged in discrete steps, not continuously.

Heisenberg's uncertainty

$$\sigma_x\,\sigma_v \;\geq\; \frac{h}{4\pi m}$$

A precise position forces an imprecise velocity, and vice versa.

Wave function

$$\int |\Psi(x,y,z,t)|^2\,dx\,dy\,dz = 1$$

Ψ is complex; $|\Psi|^2$ is the probability density of finding the particle.

Superposition and interference

$$\left|\tfrac{1}{\sqrt{2}}(\Psi_1+\Psi_2)\right|^2 = \tfrac{1}{2}\big(P_1+P_2+\underbrace{\Psi_1\Psi_2^*+\Psi_1^*\Psi_2}_{\text{interference}}\big)$$

Probabilities do not just add: the cross term is interference.

Physical qubits: photon polarisation · electron spin (±ħ/2) · trapped ions · cold atoms · superconducting circuits. Measurement and decoherence destroy superpositions.

PART 2 · A SINGLE QUBIT

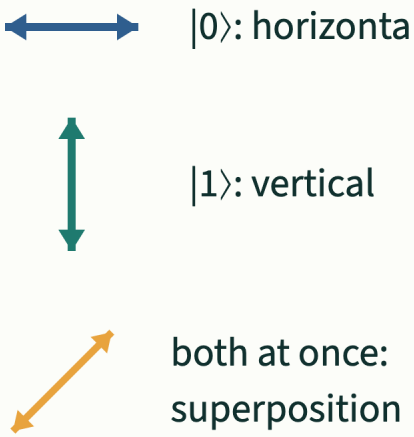
What is a qubit?

Physically: any quantum system with two distinct states, such as a photon polarised horizontally or vertically.

Mathematically: a column vector with two entries.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad |\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

The top entry says how much $|0\rangle$, the bottom how much $|1\rangle$. Both non-zero means a **superposition**.



PART 2 · A SINGLE QUBIT

Measurement and the Born rule

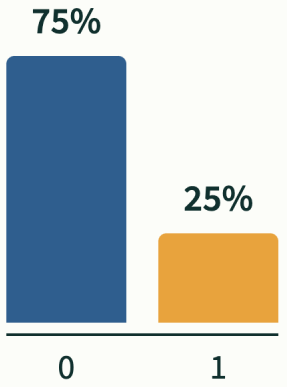
A measurement only ever returns **0 or 1**, never the amplitudes. The amplitudes give the probabilities:

$$P(0) = |\alpha|^2 \qquad P(1) = |\beta|^2$$

Afterwards the state **collapses** to the result, so every later measurement gives the same answer.

Example

$$|\psi\rangle = \begin{pmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{pmatrix}$$



$$|\alpha|^2 + |\beta|^2 = 1$$

Valid $\left(\frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}}\right)$: $\frac{1}{2} + \frac{1}{2} = 1$ **Not valid** $\left(\frac{1}{\sqrt{2}},\ 1\right)$: $\frac{1}{2} + 1 = \frac{3}{2} \neq 1$

PART 2 · A SINGLE QUBIT

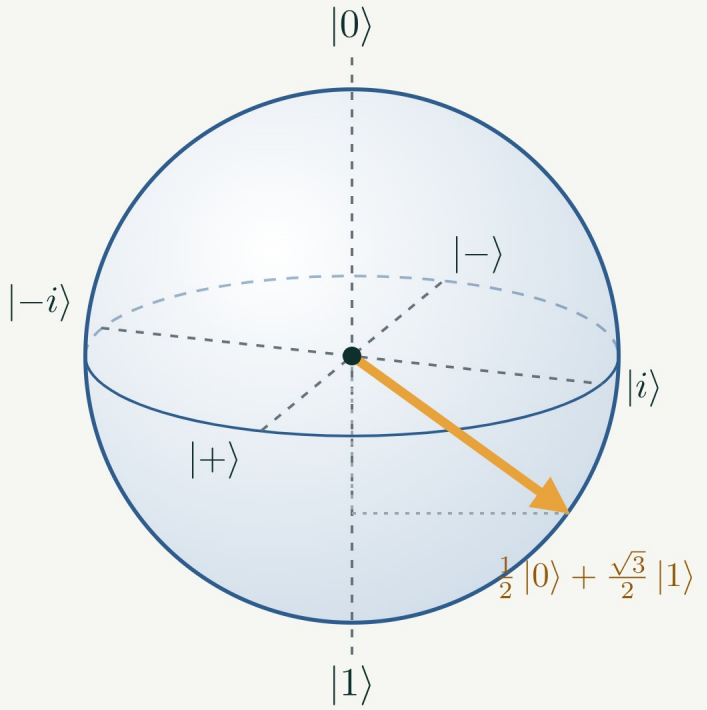
Dirac notation and the Bloch sphere

Split the vector and factor out the amplitudes:

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \alpha\,|0\rangle + \beta\,|1\rangle$$

A **ket** $|\cdot\rangle$ just marks a state. With many qubits vectors get huge, so this is the standard notation.

- North pole $|0\rangle$, south pole $|1\rangle$: the higher the arrow, the likelier 0.
- The equator holds the 50/50 states $|+\rangle$, $|-\rangle$, $|i\rangle$, $|-i\rangle$.
- Spinning around the vertical axis is **phase**.



PART 2 · A SINGLE QUBIT

Bloch sphere coordinates

Up to a global factor, every qubit can be written

$$|\psi\rangle \equiv \cos\tfrac{\theta}{2}\ |0\rangle + e^{i\varphi}\sin\tfrac{\theta}{2}\ |1\rangle$$

Its point on the unit sphere, and the chance of reading 0:

$$(x, y, z) = (\sin\theta\cos\varphi,\ \ \sin\theta\sin\varphi,\ \ \cos\theta)$$

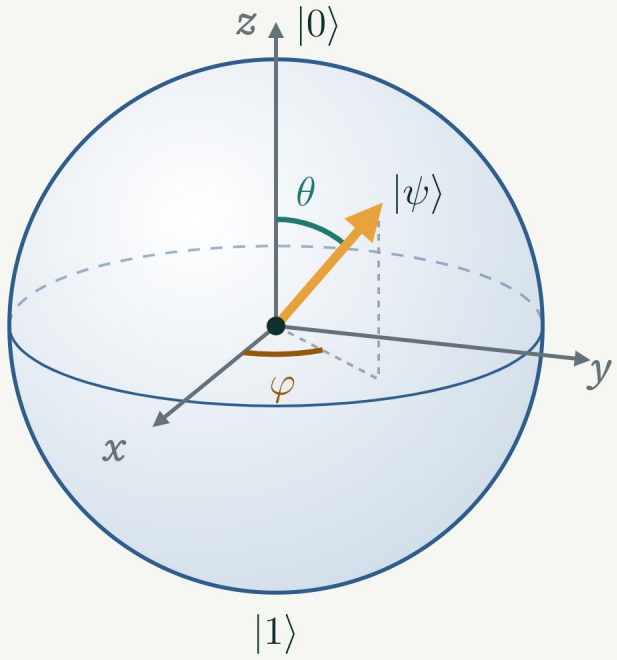
$$P(0) = \cos^2\tfrac{\theta}{2}$$

Orthogonal states sit at **opposite points**:

$$\langle\psi|\psi'\rangle = 0 \iff (\theta', \varphi') = (\pi - \theta,\ \ \varphi + \pi)$$

The Pauli gates are half-turns about the axes:

$$X:\ (\pi - \theta,\ -\varphi) \qquad Y:\ (\pi - \theta,\ \pi - \varphi) \qquad Z:\ (\theta,\ \varphi + \pi)$$



PART 2 · A SINGLE QUBIT

The Pauli gates X, Y, Z

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Half-turn about the **x**-axis: the quantum NOT.

$$X|0\rangle = |1\rangle\,, \quad X|1\rangle = |0\rangle$$

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Half-turn about the **y**-axis.

$$Y|0\rangle = i|1\rangle\,, \quad Y|1\rangle = -i|0\rangle$$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Half-turn about the **z**-axis.

$$Z(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle - \beta|1\rangle$$

Applying a gate: multiply the matrix by the state. Column 1 is where $|0\rangle$ goes, column 2 where $|1\rangle$ goes.

Linear: a gate acts on each term.

$$Y(\alpha|0\rangle + \beta|1\rangle) = -i\beta|0\rangle + i\alpha|1\rangle$$

Applying any of them twice returns the qubit: each is its **own inverse**.

PART 2 · A SINGLE QUBIT

Phase: rotating about the z-axis

$$|\psi\rangle = \alpha\,|0\rangle + e^{i\varphi}\beta\,|1\rangle$$

$Z:\ -1 = e^{i\pi},\quad |+\rangle \to |-\rangle$ a half-turn

$\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle):\ i = e^{i\pi/2}$ a quarter-turn, to |i⟩

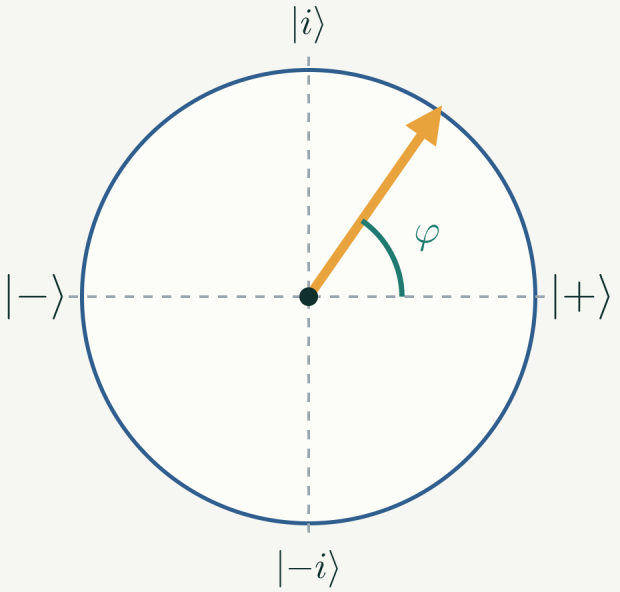
$|e^{i\varphi}\beta|^2 = |\beta|^2$ probabilities never change

Global factor

Multiplying the whole state by any non-zero complex number gives an equivalent state: no measurement can tell them apart.

Relative phase

A phase on |1⟩ only: it matters. Factor any phase on |0⟩ out as global.



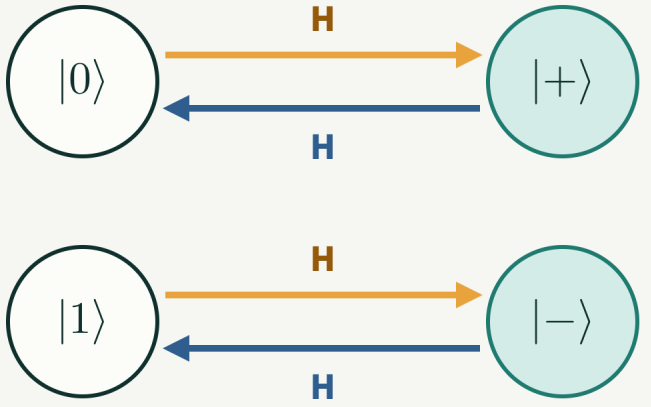
The equator, seen from above |0⟩

PART 2 · A SINGLE QUBIT

The Hadamard gate: why phase matters

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

- Its own inverse: HH = I.
- |+⟩ and |−⟩ both measure 50/50, yet H sends them to **different** states: |0⟩ and |1⟩.



Two more phase gates, with inverses S† and T†:

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \; (+\tfrac{\pi}{2}) \qquad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \; (+\tfrac{\pi}{4})$$

$\lvert+\rangle = \frac{1}{\sqrt{2}}(\lvert0\rangle + \lvert1\rangle)$	$\lvert-\rangle = \frac{1}{\sqrt{2}}(\lvert0\rangle - \lvert1\rangle)$	$\lvert i\rangle = \frac{1}{\sqrt{2}}(\lvert0\rangle + i\lvert1\rangle)$	$\lvert-i\rangle = \frac{1}{\sqrt{2}}(\lvert0\rangle - i\lvert1\rangle)$
relative phase 1	relative phase −1	relative phase i	relative phase −i

PART 3 · MULTIPLE QUBITS

Combining qubits: the tensor product

$$|0\rangle \otimes |0\rangle = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\0\\0\\0\end{pmatrix} = |00\rangle$$

Two qubits in superposition: expand like a product.

$$(\alpha\,|0\rangle + \beta\,|1\rangle) \otimes (\gamma\,|0\rangle + \delta\,|1\rangle)$$
$$= \alpha\gamma\,|00\rangle + \alpha\delta\,|01\rangle + \beta\gamma\,|10\rangle + \beta\delta\,|11\rangle$$

Two qubits: 4 basis states 00, 01, 10, 11.

Probabilities: $P(01) = |\alpha\delta|^2$

Add a qubit by tensoring it on the end.

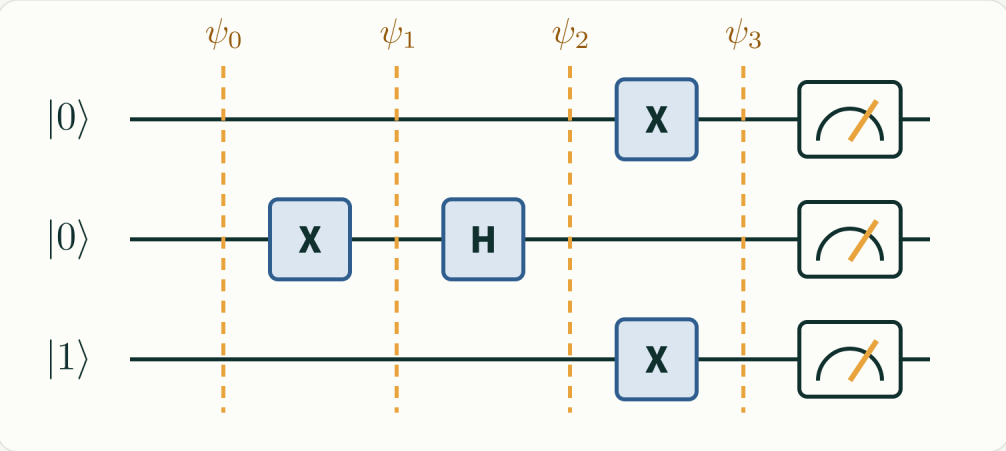
Shorthand: $|1\rangle^{\otimes 5} = |11111\rangle$

Example

$|+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$

PART 3 · MULTIPLE QUBITS

Reading a quantum circuit

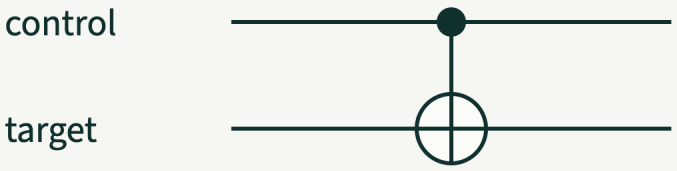


One line per qubit, gates in boxes, time flows left to right

State	Step
$\psi_0 = \|001\rangle$	initial states
$\psi_1 = \|011\rangle$	X on qubit 2
$\psi_2 = \frac{1}{\sqrt{2}}(\|001\rangle - \|011\rangle)$	H on qubit 2: $\|1\rangle \to \|-\rangle$
$\psi_3 = \frac{1}{\sqrt{2}}(\|100\rangle - \|110\rangle)$	X on qubits 1 and 3
measure	100 or 110, 50% each

PART 3 · MULTIPLE QUBITS

CNOT, Toffoli and controlled gates



In	Out
$\|00\rangle$	$\|00\rangle$
$\|01\rangle$	$\|01\rangle$
$\|10\rangle$	$\|11\rangle$
$\|11\rangle$	$\|10\rangle$

CNOT applies X to the **target** only if the **control** is $|1\rangle$. Being linear, it acts on every term of a superposition:

$$\text{CNOT}\ \tfrac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = \tfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Toffoli (CCNOT)

Two controls: flips the target only if both are $|1\rangle$. With controls 2 and 3 on target 4, $|0110\rangle \to |0111\rangle$.

Controlled-U

Any single-qubit gate can be controlled: C-Y, C-Z, C-S, C-T, C-H. Applied only when the control is $|1\rangle$.

PART 3 · MULTIPLE QUBITS

Measuring one qubit of many

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

1. Probability: add the matching terms

$$P(\text{qubit } 1 = 1) = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2}$$

$$P(\text{qubit } 2 = 1) = \tfrac{1}{4}$$

2. Collapse, then renormalise

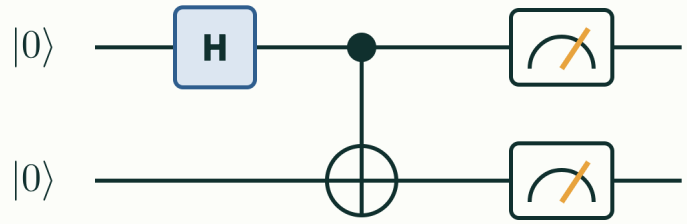
Measure qubit 1 = 1: keep the matching terms and fix the total with a constant A.

$$|A|^2\left(\tfrac{1}{4} + \tfrac{1}{4}\right) = 1 \;\Rightarrow\; A = \sqrt{2}$$

$$|\psi\rangle \to \tfrac{1}{\sqrt{2}}(|10\rangle + |11\rangle) = |1\rangle\,|+\rangle$$

PART 3 · MULTIPLE QUBITS

Entanglement



$$|00\rangle \xrightarrow{H} \tfrac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$
$$\xrightarrow{\text{CNOT}} \tfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Measure one qubit as 0 and the other collapses to 0; measure 1 and it collapses to 1.

Definition: a state is entangled if it **cannot be factored** into single-qubit states.

$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = |+\rangle\,|+\rangle$ not entangled

$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \neq |a\rangle\,|b\rangle$ **entangled**

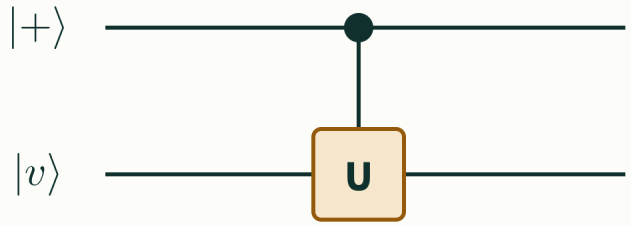
Maximally entangled: the Bell states

$$|\Phi^{\pm}\rangle = \tfrac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle) \qquad |\Psi^{\pm}\rangle = \tfrac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$$

Partially entangled: measuring one qubit only changes the other's probabilities or phase, as in the state of the previous slide.

PART 3 · MULTIPLE QUBITS

Phase kickback



$|v\rangle$ is an **eigenstate** of U:

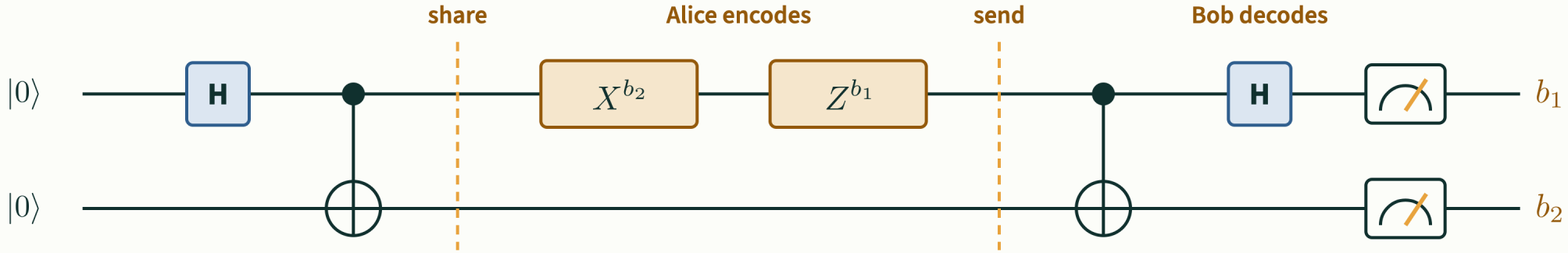
$$U\,|v\rangle = e^{i\theta}\,|v\rangle$$

$$\begin{aligned}|+\rangle\,|v\rangle &= \tfrac{1}{\sqrt{2}}\big(|0\rangle\,|v\rangle + |1\rangle\,|v\rangle\big)\\ &\to \tfrac{1}{\sqrt{2}}\big(|0\rangle\,|v\rangle + e^{i\theta}\,|1\rangle\,|v\rangle\big)\\ &= \tfrac{1}{\sqrt{2}}\big(|0\rangle + e^{i\theta}\,|1\rangle\big)\otimes|v\rangle\end{aligned}$$

The target is **unchanged**. The eigenvalue is **kicked back** as a relative phase on the control.

PART 3 · MULTIPLE QUBITS

Superdense coding: two bits for one qubit

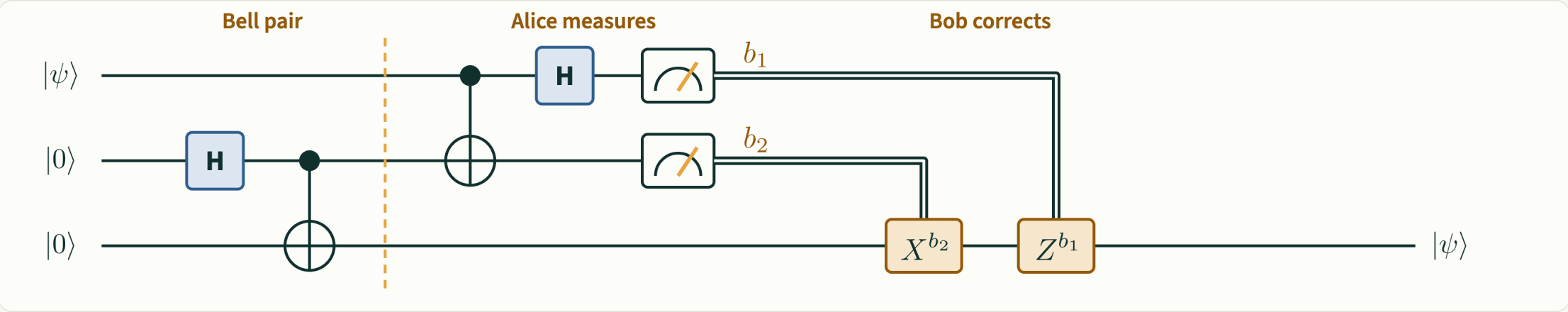


Bits	Alice applies	Shared state
00	nothing	$\|\Phi^+\rangle$
01	X	$\|\Psi^+\rangle$
10	Z	$\|\Phi^-\rangle$
11	X then Z	$\|\Psi^-\rangle$

Alice holds qubit 1, Bob qubit 2. Bob applies CNOT and H, then measures both qubits and reads Alice's **two bits**. It works because Bob already held the other half of an entangled pair.

PART 3 · MULTIPLE QUBITS

Quantum teleportation: the protocol



Not a copy

Alice's measurement destroys her qubit, so no-cloning still holds.

Not faster than light

Bob must wait for Alice's two classical bits.

Any state

α and β can be anything, unlike the 4 messages of superdense coding.

PART 3 · MULTIPLE QUBITS

Teleportation: why it works

$$|\psi\rangle \otimes |\Phi^+\rangle = \tfrac{1}{\sqrt{2}}\big(\alpha\,|000\rangle + \alpha\,|011\rangle + \beta\,|100\rangle + \beta\,|111\rangle\big)$$

$$\xrightarrow{\text{CNOT},\ H} \tfrac{1}{2}\,|00\rangle\,(\alpha\,|0\rangle + \beta\,|1\rangle) + \tfrac{1}{2}\,|01\rangle\,(\beta\,|0\rangle + \alpha\,|1\rangle)$$
$$+ \tfrac{1}{2}\,|10\rangle\,(\alpha\,|0\rangle - \beta\,|1\rangle) + \tfrac{1}{2}\,|11\rangle\,(-\beta\,|0\rangle + \alpha\,|1\rangle)$$

Alice reads **00**
$\alpha\,|0\rangle + \beta\,|1\rangle$
Bob: nothing

Alice reads **01**
$\beta\,|0\rangle + \alpha\,|1\rangle$
Bob: apply X

Alice reads **10**
$\alpha\,|0\rangle - \beta\,|1\rangle$
Bob: apply Z

Alice reads **11**
$-\beta\,|0\rangle + \alpha\,|1\rangle$
Bob: X, then Z

Equivalently, Alice measures her two qubits in the **Bell basis**:

$$|\psi\rangle\,|\Phi^+\rangle = \tfrac{1}{2}\big(|\Phi^+\rangle\,|\psi\rangle + |\Psi^+\rangle\,X\,|\psi\rangle + |\Phi^-\rangle\,Z\,|\psi\rangle + |\Psi^-\rangle\,XZ\,|\psi\rangle\big)$$

PART 3 · MULTIPLE QUBITS

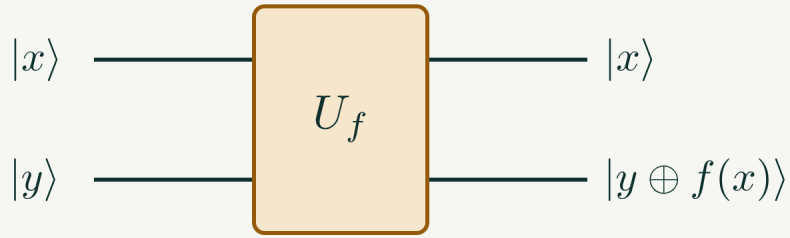
Reversible functions and quantum oracles

Every quantum operation must be reversible

Gates are unitary, so only measurement may lose information. But OR is not reversible: an output of 1 could come from three inputs.

Fix: keep the input and XOR the output into an extra bit c.

$$(x, c) \mapsto (x,\ c \oplus f(x))$$



With y = 0 you get |x⟩|f(x)⟩: measure the second register to read f.

PHASE ORACLE: SET Y = |−⟩

$$U_f\,|x\rangle\,|-\rangle = (-1)^{f(x)}\,|x\rangle\,|-\rangle$$

The answer becomes a phase. Deutsch, Deutsch–Jozsa and Bernstein–Vazirani all rely on it.

PART 3 · MULTIPLE QUBITS

The no-cloning theorem

No gate can copy an unknown qubit.

- The proof uses only **linearity**, no measurement.
- A *known* state can be prepared again with the right gates.
- A CNOT “fan-out” gives α|00⟩ + β|11⟩: entangled, not a copy.
- The same argument works for n qubits.

Suppose a gate A could do

$$A\ |\psi\rangle\,|0\rangle = |\psi\rangle\,|\psi\rangle \quad \text{for every} \ \ |\psi\rangle\,?$$

1. It copies the basis states:

$$A\,|00\rangle = |00\rangle\,, \qquad A\,|10\rangle = |11\rangle$$

2. So, by linearity, on |+⟩ it must give

$$A\,|+\rangle\,|0\rangle = \tfrac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big)$$

3. But a copy of |+⟩ is

$$|+\rangle\,|+\rangle = \tfrac{1}{2}\big(|00\rangle + |01\rangle + |10\rangle + |11\rangle\big)$$

The |01⟩ terms differ (0 versus ½): contradiction.

PART 3 · MULTIPLE QUBITS

Universal gate sets

Classical logic: Toffoli is universal

$$|x\rangle\,|y\rangle\,|1\rangle \;\mapsto\; |x\rangle\,|y\rangle\,|1 \oplus xy\rangle$$

With its target set to 1 it computes NAND, and NAND builds every Boolean function.

Exact: CNOT plus one-qubit gates

$$U(\theta,\varphi,\lambda) = \begin{pmatrix} \cos\frac{\theta}{2} & -e^{i\lambda}\sin\frac{\theta}{2} \\ e^{i\varphi}\sin\frac{\theta}{2} & e^{i(\varphi+\lambda)}\cos\frac{\theta}{2} \end{pmatrix}$$

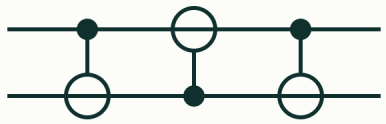
Every n-qubit gate is a circuit of CNOTs and one-qubit gates like this one.

Approximate: just three gates

$$H, \qquad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}, \qquad \mathrm{CNOT}$$

They approximate any gate as closely as you like, with few gates (Solovay–Kitaev).

Two useful facts



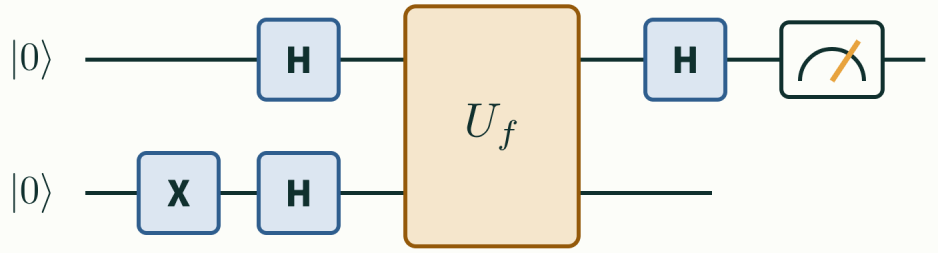
= SWAP

$$\alpha\,|0\rangle + \beta\,|1\rangle \;\mapsto\; \alpha\,|00\rangle + \beta\,|11\rangle \;\neq\; (\alpha\,|0\rangle + \beta\,|1\rangle)^{\otimes 2}$$

PART 4 · ALGORITHMS

Deutsch's algorithm: constant or balanced?

f	f(0)	f(1)	Type
const 0	0	0	constant
const 1	1	1	constant
identity	0	1	balanced
NOT	1	0	balanced



Classically: query f(0) *and* f(1), so **2 queries**. Quantum: **1 query**.

$$|+\rangle|-\rangle \xrightarrow{U_f} \frac{1}{\sqrt{2}}\left((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\right)|-\rangle = \begin{cases} \pm|+\rangle|-\rangle & f(0)=f(1) \\ \pm|-\rangle|-\rangle & f(0)\neq f(1) \end{cases} \xrightarrow{H} \pm|0\rangle \text{ or } \pm|1\rangle$$

Measure 0 ⇒ constant. Measure 1 ⇒ balanced. One query.

PART 4 · ALGORITHMS

Deutsch–Jozsa: n-bit inputs, still one query

$2^{n-1}+1$ classical queries in the worst case: half the inputs agree, so one more is needed

1 quantum query: Hadamards, the phase oracle, Hadamards, then measure

Two identities that keep coming back

$$H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}}\sum_x |x\rangle$$

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}}\sum_z (-1)^{x\cdot z}|z\rangle$$

A uniform superposition of every bit string; x·z is the bitwise dot product mod 2.

Why it works: the amplitude of |0…0⟩

$$a_{0\cdots 0} = \frac{1}{2^n}\sum_x (-1)^{f(x)}$$

Constant: every term is +1 (or −1), amplitude ±1, so you always read 0…0.
Balanced: +1s and −1s cancel, amplitude 0, so you never do.

PART 4 · ALGORITHMS

Bernstein–Vazirani: find a secret string

$$f(x) = s \cdot x \bmod 2$$

- **Classical**: query 100…, 010…, … to learn one bit of s each time: n queries.
- **Quantum**: the Deutsch–Jozsa circuit with this oracle: 1 query.

After the oracle, then the final Hadamards:

$$\frac{1}{\sqrt{2^n}} \sum_x (-1)^{s \cdot x} |x\rangle$$

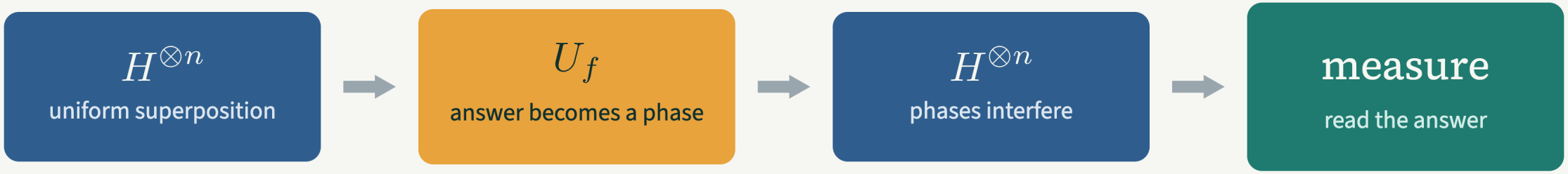
$$\xrightarrow{H^{\otimes n}} \frac{1}{2^n} \sum_z \sum_x (-1)^{(s \oplus z) \cdot x} |z\rangle$$

For z = s every exponent is 0, so all the terms are +1 and the amplitude of |s⟩ is exactly 1.

Measuring returns s with certainty.

PART 4 · ALGORITHMS

The common pattern

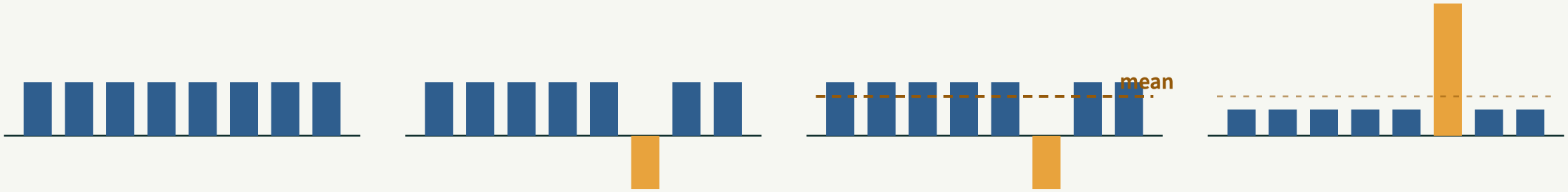


Deutsch, Deutsch–Jozsa and Bernstein–Vazirani share this skeleton. Next come the **quantum Fourier transform** and **phase estimation**, the building blocks of Shor's algorithm.

PART 4 · ALGORITHMS

Grover's search: the idea

Find the one k_0 with $f(k_0) = 1$ in an **unsorted** list of N items. Each round: flip the sign of the answer, then reflect every amplitude about the mean.



1 · Equal amplitudes: no information yet

2 · The oracle flips the answer's sign

3 · Compute the mean

4 · Reflect about the mean: the answer grows

Sorted list (binary search)

about $\log_2 N$ tests

Unsorted, classical

up to N tests

Unsorted, Grover

about $(\pi/4)\cdot\sqrt{N}$ rounds

PART 4 · ALGORITHMS

Grover's geometry: two reflections make a rotation

Split the uniform state into the answer $|k_0\rangle$ and the rest $|\chi\rangle$:

$$|\psi_H\rangle = H^{\otimes n}\,|0\cdots 0\rangle = \cos\tfrac{\theta}{2}\;|\chi\rangle + \sin\tfrac{\theta}{2}\;|k_0\rangle\,, \qquad \sin\tfrac{\theta}{2} = \tfrac{1}{\sqrt{N}}$$

One round: the oracle, then a reflection about $|\psi_H\rangle$:

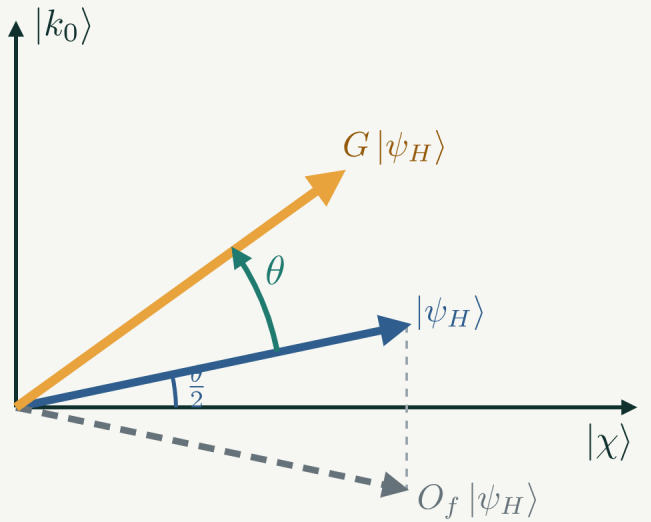
$$G = S_{\psi_H}\,O_f, \qquad S_{\psi_H} = 2\,|\psi_H\rangle\,\langle\psi_H| - I = H^{\otimes n} S_0\,H^{\otimes n}$$

Each round rotates the state by θ towards $|k_0\rangle$:

$$G^{\ell}\,|\psi_H\rangle = \cos\theta_\ell\;|\chi\rangle + \sin\theta_\ell\;|k_0\rangle\,, \qquad \theta_\ell = \tfrac{\theta}{2} + \ell\theta$$

Stop when $\theta_\ell \approx \pi/2$, then measure:

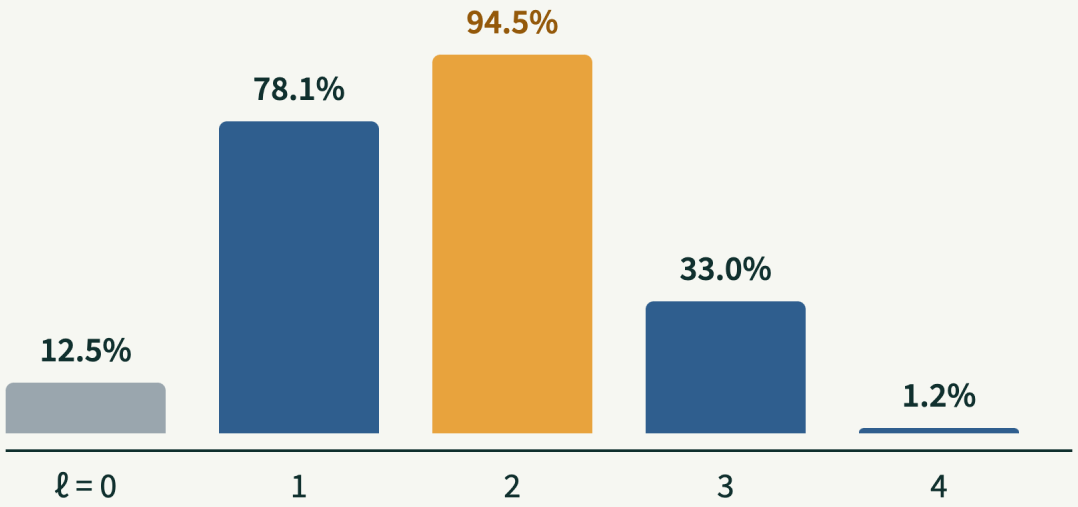
$$\ell \approx \tfrac{\pi}{4}\sqrt{N}, \qquad P(\text{success}) \geq 1 - \tfrac{4}{N}$$



PART 4 · ALGORITHMS

Grover in practice

N = 8, answer |101⟩: chance of success after ℓ rounds



Two rounds is best, matching $(\pi/4)\cdot\sqrt{8} \approx 2.2$. Going further **overshoots**: the rotation carries the state past $|k_0\rangle$.

Inverse problems

Any “find x with f(x) = y” search where the only method is to try every x.

Brute-forcing keys

$$2^n \rightarrow 2^{n/2}$$

An n-bit key falls in 2^(n/2) steps, hence the advice to move from AES-128 to AES-256.

Probabilistic

Usually right; check the answer and rerun if it is wrong.

PART 4 · ALGORITHMS

Quantum Fourier transform: numbers in phases

In bits, 5 = 101. After the QFT every qubit sits on the equator and the number lives in the **phases**, which turn by

$$2\pi\,\frac{j}{2}, \quad 2\pi\,\frac{j}{4}, \quad 2\pi\,\frac{j}{8}$$

$$\mathrm{QFT}\,|j\rangle = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i\, jk/2^n}\,|k\rangle$$

Circuit: H gates, controlled rotations, then swaps.

$$R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{pmatrix}$$

QFT|101⟩, each qubit seen from above |0⟩



$2\pi \cdot \frac{5}{2} \equiv \pi$ $\qquad$ $2\pi \cdot \frac{5}{4} \equiv \frac{\pi}{2}$ $\qquad$ $2\pi \cdot \frac{5}{8} = \frac{5\pi}{4}$

PART 4 · ALGORITHMS

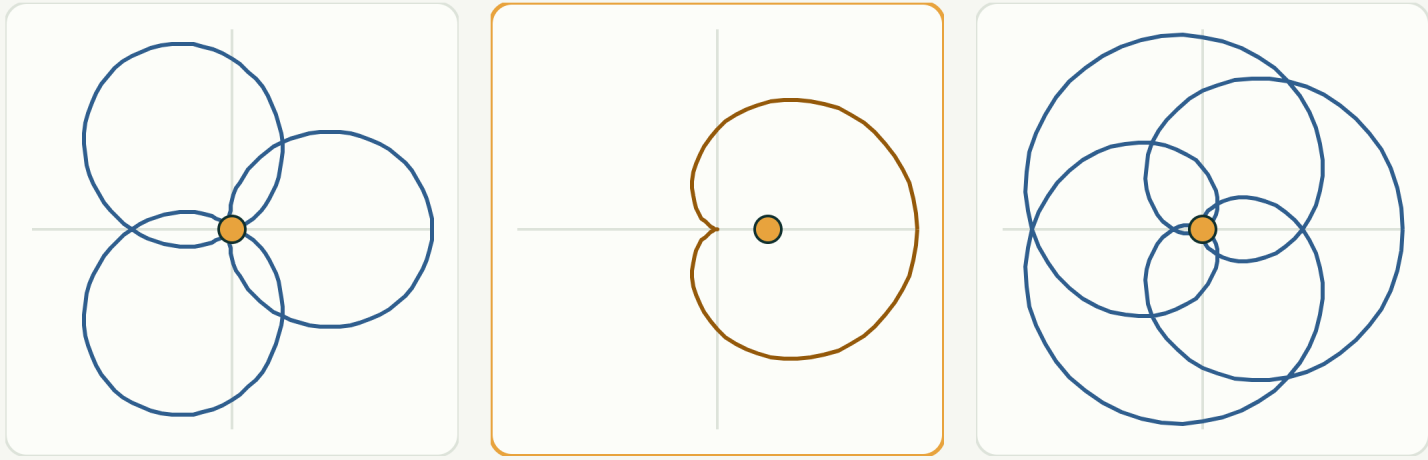
Why Fourier finds periods: winding

Wrap the data around a circle: point j at distance x_j, turned by t full turns over the whole list. The centre of mass is

$$X_t = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} x_j \, e^{-2i\pi \, tj/n}$$

For whole t this is the discrete Fourier transform; the QFT does the same to amplitudes.

> Only when t matches the signal's frequency does the centre of mass leave the origin.



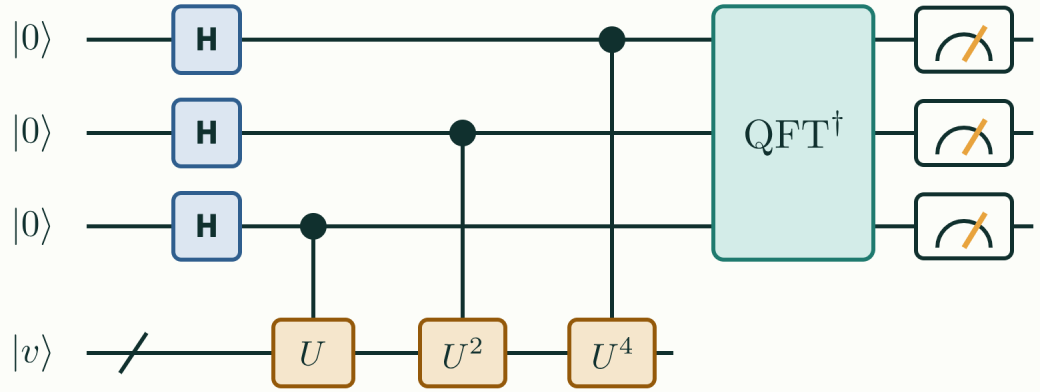
$t = 2$ $t = 3$ $t = 4$

$f(x) = 1 + \cos(3x)$ wound t times around the circle; the dot is the centre of mass.

PART 4 · ALGORITHMS

Quantum phase estimation

Goal: given U and an eigenvector |v⟩, find θ to m bits, where $U|v\rangle = e^{i\theta}|v\rangle\,, \qquad \theta = 2\pi j$



1. $H^{\otimes m}: |0\rangle^{\otimes m} \to |+\rangle^{\otimes m}$
2. $\text{C-}U^{2^k}$: qubit k picks up $e^{i\,2^k\theta}$
3. $j = 0.j_1 j_2 \ldots j_m \;\Rightarrow\; \text{register} = \text{QFT}\,|j_1 \ldots j_m\rangle$
4. $\text{QFT}^\dagger \to |j_1 \ldots j_m\rangle\,, \qquad \theta = 2\pi j$

Step 2 is phase kickback.

PART 4 · ALGORITHMS

Why factoring matters: RSA

1 · ALICE, IN SECRET

Picks two large primes p and q, then e, and computes d.

2 · PUBLIC KEY

Alice publishes (n, e). Anyone can encrypt a message for her.

3 · PRIVATE KEY

Only Alice knows d, so only she can decrypt.

$$n = pq, \qquad \varphi(n) = (p-1)(q-1), \qquad d\,e \equiv 1 \pmod{\varphi(n)}$$

$$x \equiv m^{e} \pmod{n} \qquad\qquad m \equiv x^{d} \pmod{n}$$

Example: $n = 55,\ \varphi(n) = 40,\ e = 3,\ d = 27:\qquad 41^{3} \equiv 6, \qquad 6^{27} \equiv 41 \pmod{55}$

> Knowing p and q gives ϕ(n), hence d. So whoever can **factor n** can read every message. Today's keys use 2048 bits; the classical factoring record is about 250 digits (800 bits).

PART 4 · ALGORITHMS

Shor's algorithm: factoring N = pq

- Factoring underpins **RSA** encryption.
- Shor reduces factoring to **period finding**, which a quantum computer does efficiently.
- Breaking RSA-2048 in hours was estimated at about **20 million physical qubits** (figure in the Exo7 book); a 2025 estimate brings it under 1 million, in under a week.

1 · Classical $1 < a < N; \quad \gcd(a, N) \neq 1 \Rightarrow \text{done}$

⬇

2 · Quantum $\text{period } r \text{ of } a^x \bmod N$

⬇

3 · Classical $\text{continued fractions} \to r$

⬇

4 · Classical $\gcd\left(a^{r/2} \pm 1,\ N\right)$

PART 4 · ALGORITHMS

Modular exponentiation is periodic

a ≡ b (mod N) means a and b leave the same remainder when divided by N, e.g. 3 ≡ 1 (mod 2).

x	0	1	2	3	4	5	6	7	8
$2^x \bmod 21$	1	2	4	8	16	11	1	2	4

$$r = 6$$

The sequence starts at $a^0 = 1$ and returns to 1 after r steps.

$$(a^{r/2} - 1)(a^{r/2} + 1) \equiv 0 \pmod{N}$$

Since $a^r \equiv 1 \pmod{N}$: a difference of two squares.

$$\gcd(7, 21) = 7, \qquad \gcd(9, 21) = 3$$

$$\gcd(a^{r/2} \pm 1,\ N)$$

likely gives the factors.

PART 4 · ALGORITHMS

Every choice of a for N = 15

2 r = 4 3 × 5	**3** gcd = 3 done	**4** r = 2 3 × 5	**5** gcd = 5 done	**6** gcd = 3 done	**7** r = 4 3 × 5	**8** r = 4 3 × 5
9 gcd = 3 done	**10** gcd = 5 done	**11** r = 2 5 × 3	**12** gcd = 3 done	**13** r = 4 3 × 5	**14** 14 ≡ −1 fails	**teal**: order works **blue**: gcd shortcut **amber**: fails

The order trick works when

$$r \text{ even} \quad \text{and} \quad a^{r/2} \not\equiv -1 \pmod{N}$$

With ℓ distinct prime factors ($\ell = 2$ for RSA):

$$P(a \text{ works}) \geq 1 - \frac{1}{2^{\ell-1}}$$

PART 4 · ALGORITHMS

The quantum step: period finding is phase estimation

The gate U|x⟩ = |ax mod N⟩

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i\, sk/r}\, |a^k \bmod N\rangle$$

$$U\,|u_s\rangle = e^{2\pi i\, s/r}\,|u_s\rangle$$

The eigenvalue contains the period r.

The trick

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle$$

|1⟩ is easy to prepare. Phase estimation on |1⟩ returns s/r for a random s. If s = 0, run again.

Continued fractions

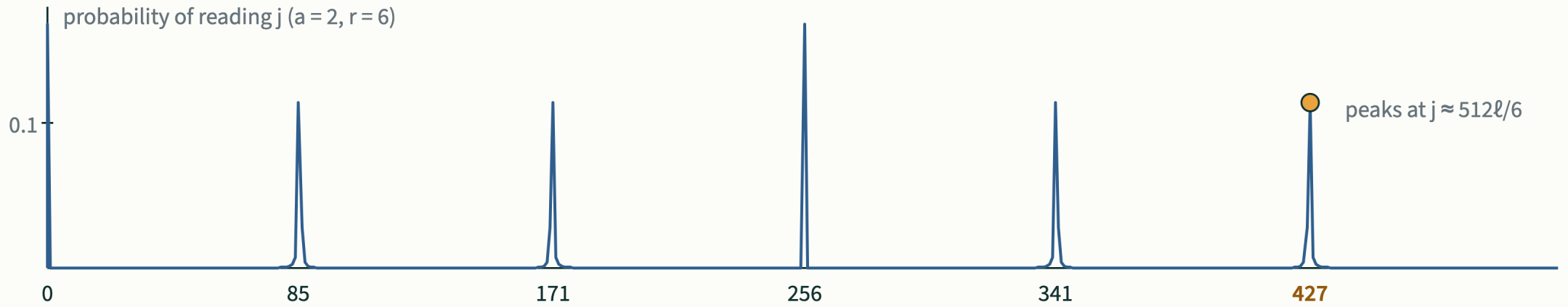
$$0.312 = \frac{39}{125} = \cfrac{1}{3 + \cfrac{1}{4 + \cfrac{1}{1 + \cfrac{1}{7}}}}$$

Truncations: $\frac{1}{3}, \quad \frac{4}{13}, \quad \frac{5}{16}, \ \ldots$

Keep the one with denominator below N: that is r. If r is odd, repeat.

PART 4 · ALGORITHMS

What the measurement looks like: N = 21



Size the counting register so that

$N^2 \le 2^n < 2N^2$: $\quad 441 \le 512 < 882 \;\Rightarrow\; n = 9$

$$\left|\frac{j}{2^n} - \frac{\ell}{r}\right| \le \frac{1}{2^{n+1}} < \frac{1}{2r^2} \;\Rightarrow\; \frac{\ell}{r} \text{ is a convergent of } \frac{j}{2^n}$$

Reading the orange peak:

$$\frac{427}{512} = [0; 1, 5, 42, 2] \;\to\; \frac{1}{1},\ \frac{5}{6},\ \ldots \;\Rightarrow\; r = 6$$

Peaks 171 or 341 only give 3, a factor of r; 256 gives 2; 0 gives nothing: run again.

PART 4 · ALGORITHMS

Worked example: factoring 15

STEP 1

Choose **a = 7**, coprime to 15.

STEP 2 · QUANTUM PERIOD FINDING

x	0	1	2	3	4
$7^x \bmod 15$	1	7	4	13	1

Period **r = 4**.

STEP 3 · GCDS WITH $7^2 = 49$

$$\gcd(48, 15) = 3$$

$$\gcd(50, 15) = 5$$

$15 = 3 \times 5$

What the counting register actually shows

With 3 counting qubits, phase estimation reads 000, 010, 100 or 110: $\frac{s}{r} \in \left\{0,\ \frac{1}{4},\ \frac{1}{2},\ \frac{3}{4}\right\}$

A quarter or three quarters gives r = 4. A half suggests r = 2, but $7^2 \bmod 15 = 4 \neq 1$ so run again.

PART 5 · LIVING IN A QUANTUM WORLD

Quantum cryptography: secret keys from physics

The one-time pad is perfectly secure

$$y = x \oplus m, \qquad y \oplus m = x$$

```
message x   1011011
key     m   0010111
sent    y   1001100
```

The key must be random, secret, as long as the message and used once. The hard part is **sharing it**.

BB84 (Bennett & Brassard, 1984): two bases

Qubit sent	$\|0\rangle$	$\|1\rangle$	$\|+\rangle$	$\|-\rangle$
Bit	0	1	0	1
Read in ⊕	0	1	random	random
Read in ⊗	random	random	0	1

Right basis: the exact bit. Wrong basis: a coin flip. ⊗ is ⊕ with a Hadamard before sending and before measuring.

The two encodings: $\oplus$: $|0\rangle \to 0, \quad |1\rangle \to 1$ $\qquad$ $\otimes$: $|+\rangle = H|0\rangle \to 0, \quad |-\rangle = H|1\rangle \to 1$

PART 5 · LIVING IN A QUANTUM WORLD

BB84: sharing a key and catching Eve

Position	1	2	3	4	5	6	7	8	9
Alice's bit	1	0	0	1	1	1	0	1	0
Alice's basis	⊕	⊗	⊕	⊗	⊗	⊕	⊕	⊗	⊗
Bob's basis	⊕	⊕	⊕	⊗	⊕	⊗	⊕	⊗	⊕
Shared key bit	1	–	0	1	–	–	0	1	–

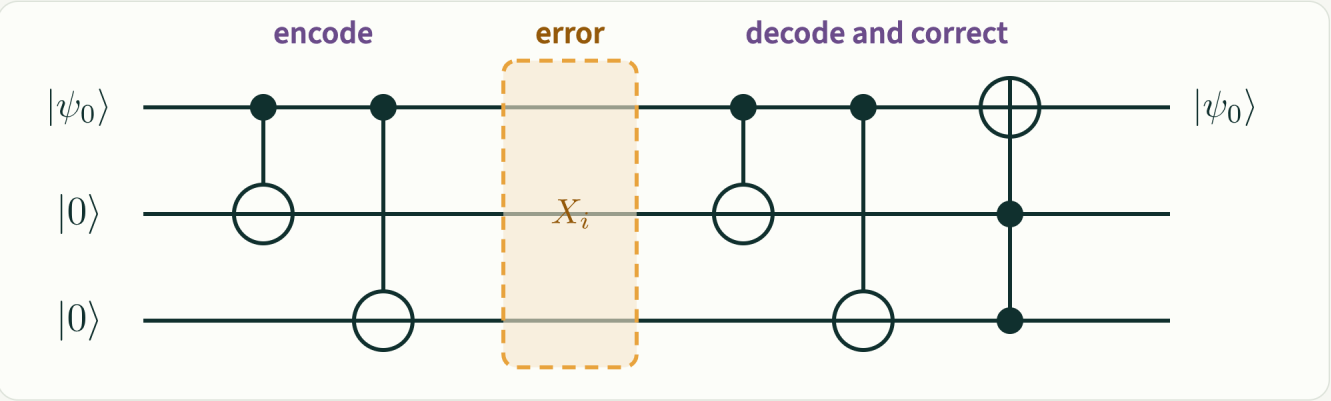
1. Alice sends random bits, each in a random basis.
2. Bob measures each qubit in a random basis.
3. They compare **bases** in public and keep the positions where they match: here 1 0 1 0 1.
4. They reveal a sample of n kept bits: any mismatch means a spy.

Eve can't copy qubits (no-cloning). Guessing the basis, she corrupts each kept bit with probability ¼, so

$$P(\text{Eve unnoticed}) = \left(\tfrac{3}{4}\right)^n$$

n = 10: 5.6% · n = 20: 0.3% · n = 100: 3×10^{-13}

Error correction: the bit-flip code



Three obstacles

- Measuring disturbs the qubit.
- Qubits cannot be cloned.
- α and β are continuous, not just 0 or 1.

And yet errors can be corrected.

Classical idea

Send 0 as 000 and 1 as 111, then take a majority vote.

Encode by entangling, not copying

$$\alpha|0\rangle + \beta|1\rangle \mapsto \alpha|000\rangle + \beta|111\rangle$$

A flip on line 1 is undone

$$\alpha|100\rangle + \beta|011\rangle \xrightarrow{\text{decode}} (\alpha|0\rangle + \beta|1\rangle)|11\rangle$$

PART 5 · LIVING IN A QUANTUM WORLD

Phase flips, any error, Shor's code

Phase flips

$$HZH = X$$

Z swaps $|+\rangle$ and $|-\rangle$: a bit flip in another basis. Wrap the same code in Hadamards and it corrects Z errors.

Any single-qubit error

$$E = aI + bX + cY + dZ, \qquad Y = i\,XZ$$

Correct X and Z and you correct everything: measuring the helper qubits **discretises** the error.

Shor's code: 9 qubits for 1

$$|0_L\rangle = \tfrac{1}{2\sqrt{2}}\big(|000\rangle + |111\rangle\big)^{\otimes 3}, \qquad |1_L\rangle = \tfrac{1}{2\sqrt{2}}\big(|000\rangle - |111\rangle\big)^{\otimes 3}$$

The bit-flip code nested inside the phase-flip code. It corrects any error on any one of its nine lines.

PART 5 · LIVING IN A QUANTUM WORLD

Quantum advantage: when does quantum win?

$2^{50} \approx 10^{15}$

amplitudes to store to simulate just 50 qubits

≈ 250 digits

classical factoring record (about 800 bits)

2048 bits

recommended RSA key size

What “advantage” means

A quantum computer solves a problem that no classical computer can solve in reasonable time. Proving no classical algorithm exists is hard. “Advantage” is now preferred to “supremacy”: the two kinds of machine will work together.

Not every circuit is hard (Gottesman–Knill)

Circuits made only of H, S, CNOT and Pauli gates, started from $|0\rangle$, are simulated efficiently on a laptop: superdense coding, teleportation, error detection. Add T or Toffoli and they become universal.

Figures as given in the Exo7 book (2020): quantum machines then had about 50 qubits and had factored numbers of 5 digits.

PART 5 · LIVING IN A QUANTUM WORLD

Try it with myQLM: a Bell pair

```
from qat.lang.AQASM import Program, H, CNOT
from qat.qpus import PyLinalg

prog = Program()
q = prog.qalloc(2)
prog.apply(H, q[0])
prog.apply(CNOT, q[0], q[1])
circ = prog.to_circ()

result = PyLinalg().submit(circ.to_job())
for sample in result:
    print(sample.state, sample.probability)
```

OUTPUT

```
|00> 0.4999999999999999
|11> 0.4999999999999999
```

With to_job(nbshots=1000), one run gave |00> 0.484 and |11> 0.516.

- Install: **pip install myqlm**
- Same circuit as the entanglement slide: H, then CNOT.
- PyLinalg is myQLM's pure-Python simulator; get_default_qpu() picks a faster C one that needs OpenMP.

Grover with myQLM: finding |101⟩

```
from qat.lang.AQASM import Program, H, X, Z
from qat.qpus import PyLinalg
prog = Program()
q = prog.qalloc(3)
CCZ = Z.ctrl().ctrl()                    # doubly controlled Z
for k in range(3): prog.apply(H, q[k])
for _ in range(2):                       # about (pi/4)*sqrt(8) rounds
    prog.apply(X, q[1])                  # oracle: flip the sign of |101>
    prog.apply(CCZ, q[0], q[1], q[2])
    prog.apply(X, q[1])
    for k in range(3): prog.apply(H, q[k]); prog.apply(X, q[k])
    prog.apply(CCZ, q[0], q[1], q[2])  # diffusion: reflect about the mean
    for k in range(3): prog.apply(X, q[k]); prog.apply(H, q[k])
result = PyLinalg().submit(prog.to_circ().to_job())
for s in result: print(s.state, round(s.probability, 4))
```

OUTPUT



0.9453 = 121/128, exactly the theory of the Grover slides.

SUMMARY

Takeaways

1. A qubit is a unit complex vector, a point on the Bloch sphere up to a global factor. Measuring gives $|\alpha|^2$, $|\beta|^2$ and collapses it.

2. Gates are **unitary**: they keep inner products. Relative phase is invisible to measurement but changes what gates do.

3. **Entanglement** powers superdense coding and teleportation; linearity forbids cloning; H, T and CNOT are enough for everything.

4. Interference does the work: one query for Deutsch–Jozsa, about $\sqrt{N}$ rounds for **Grover**.

5. **Shor** turns factoring into period finding with the QFT; it works for at least half the choices of a, and would break RSA.

6. Physics also protects: **BB84** exposes spies, error-correcting codes fix any one-qubit error, and you can try it all in myQLM.

From $i = \sqrt{-1}$ to encryption protocol

Sources: CERFACS - BULL Training - October 2026